



Modelling Financial Markets

Chapter 7: Value-at-Risk and Expected Shortfall



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Acronyms used in this material

- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GEV: Generalised Extreme Value (distribution)
- ▣ GPD: Generalised Pareto Distribution
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

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How Much Can We Lose?

The question of this chapter

- A bank, a fund or an investor holds a portfolio today; its value tomorrow is random
- Management needs **one number**: a loss that is exceeded only rarely
- Regulators need a number to set **capital**, the buffer that absorbs losses

What we do in this chapter

- Define Value-at-Risk (VaR) and Expected Shortfall and their properties
- Estimate them with historical, parametric, filtered and extreme-value methods
- Scale them to longer horizons and split them across the assets of a portfolio
- Connect them to the Basel capital rules

RiskMetrics: VaR Becomes an Industry Standard



J.P. Morgan & Co. building, 23 Wall Street, New York

Photo: Beyond My Ken (2012); CC BY-SA 4.0; Wikimedia Commons

- J.P. Morgan launched **RiskMetrics** in October 1994: a free methodology and data set for measuring market risk [J.P. Morgan/Reuters \(1996\)](#)
 - ▶ Volatilities and correlations for many markets, updated daily
- The fourth edition of the Technical Document (1996) became the reference text for VaR
- In the same year the Basel Committee let banks use internal VaR models for capital [Basel Committee \(1996\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#), [Dánielsson \(2011\)](#)

Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	2000–2026	6 714
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
EUR/RON	BNR reference rate	2005–2026	5 352
Gold	USD per troy ounce, weekdays	2000–2026	6 956
Portfolio	SPY, TLT, GLD (ETFs) and Bitcoin	2014–2026	3 017

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$, in %
- Each series on its own calendar; the portfolio uses the days on which all four prices exist
- ETFs: prices adjusted for dividends; SPY tracks the S&P 500, TLT long US Treasury bonds, GLD gold

Losses, Horizons and Tail Probabilities

- Return (P&L) over one day: $X_{t+1} = r_{t+1}$, in % of the position; **loss** $L_{t+1} = -X_{t+1}$
 - ▶ Positive L = the position lost money; negative L = a gain
 - ▶ In money: $\text{loss} = V_t \times L_{t+1}/100$, with V_t the value of the position
- **Horizon** h : the number of days over which the loss is measured (1 day, 10 days)
- **Level** α : the **tail probability**, typically $\alpha = 1\%$ or 2.5% : “VaR 1%”, “ES 2.5%”
 - ▶ α is the probability of a worse outcome than the risk measure
- **Sign convention in this course**: VaR and ES are **positive** numbers when they are losses

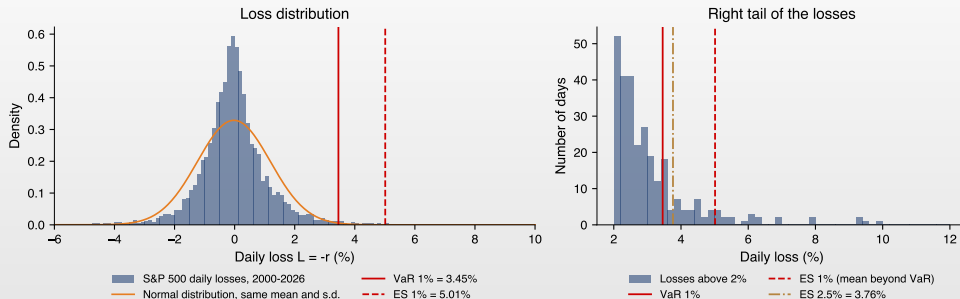
Definition: Value-at-Risk

- **VaR** at level α : the loss that is exceeded with probability α
 - ▶ $\text{VaR}_\alpha(X) = -\inf\{x \in \mathbb{R} : P(X \leq x) > \alpha\} = -q_\alpha(X)$, minus the α -quantile of the return
 - ▶ Equivalently, for the loss $L = -X$ with a continuous distribution: $\text{VaR}_\alpha = q_{1-\alpha}(L)$
- Reading of VaR 1%: “with probability 1%, tomorrow’s loss will exceed $\text{VaR}_{1\%}$ ”
- Example: a one-day VaR 1% of 3.45% on 1 million EUR: on 1 day in 100 we expect to lose more than 34 547 EUR
- What VaR does **not** say: how large the loss is when it exceeds VaR
 - ▶ Two portfolios with the same VaR can have very different tails

Definition: Expected Shortfall

- **Expected Shortfall:** the average of all quantiles in the tail of probability α
 - ▶
$$\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = \frac{1}{\alpha} \int_0^\alpha \text{VaR}_u(X) du$$
- For a continuous distribution: $\text{ES}_\alpha(X) = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$, the mean loss on the worst α share of days
- Always $\text{ES}_\alpha \geq \text{VaR}_\alpha$: ES looks **into** the tail, VaR only at its entrance
- Other names: Conditional VaR (CVaR), Tail VaR, Average VaR [Rockafellar & Uryasev \(2002\)](#), [Acerbi & Tasche \(2002\)](#)
- Historical estimate: the mean of the observed losses at or above the empirical VaR

VaR and ES on the S&P 500



□ Left: daily losses 2000–2026 and the Normal distribution with the same mean and standard deviation; right: the tail

Reading the Loss Distribution

- ▣ Historical $\text{VaR}_{1\%} = 3.45\%$: 68 of 6 714 days (1.01%) had a larger loss
 - ▶ $\text{ES}_{1\%} = 5.01\%$: on those days the average loss was 5.01%
- ▣ $\text{VaR}_{2.5\%} = 2.51\%$ and $\text{ES}_{2.5\%} = 3.76\%$: ES 2.5% exceeds VaR 1%
- ▣ The Normal distribution fits the centre but is far too thin in the tail (Chapter 1): heavy tails
- ▣ Largest loss: 12.77% on 2020-03-16 (COVID-19 crash), about ten standard deviations

Worked Example: the Normal Distribution

- If $X \sim N(\mu, \sigma^2)$ and $z_\alpha = \Phi^{-1}(\alpha) < 0$, with Φ, φ the standard Normal distribution and density functions
 - ▶ $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha) = -\mu + \sigma z_{1-\alpha}, \quad \text{ES}_\alpha = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha}$
- Constants: $z_{1\%} = -2.326, \varphi(z_{1\%})/0.01 = 2.665; z_{2.5\%} = -1.960, \varphi(z_{2.5\%})/0.025 = 2.338$
- S&P 500: mean daily return $\hat{\mu} = 0.025\%, \hat{\sigma} = 1.21\%$
 - ▶ $\text{VaR}_{1\%} = 2.80\%, \text{ES}_{1\%} = 3.21\%$ versus historical 3.45% and 5.01%
- The Normal VaR is too low by a factor of 1.24: heavy tails matter at $\alpha = 1\%$

Worked Example: the Student- t Distribution

- $X = m + s T$ with $T \sim t_\nu$ (location m , scale s , degrees of freedom ν); $t_\alpha = t_\nu^{-1}(\alpha) < 0$, g_ν the density
 - ▶ $\text{VaR}_\alpha = -(m + s t_\alpha)$, $\text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$ (for $\nu > 1$)
- Variance $s^2\nu/(\nu - 2)$ exists only for $\nu > 2$; smaller ν = heavier tails
- Maximum likelihood fit to S&P 500 returns: $\hat{\nu} = 2.67$
 - ▶ $\text{VaR}_{1\%} = 3.46\%$ (historical 3.45%), $\text{ES}_{1\%} = 5.70\%$ (historical 5.01%)
- The t distribution fixes the VaR, but with $\hat{\nu} < 3$ it overstates ES: one parameter must fit both the centre and the tail

Recap: Definitions

- ▣ VaR 1% is the loss exceeded with probability 1%; ES 1% is the average loss in that tail
- ▣ Both depend on the horizon h , the tail probability α and the model of the return distribution
- ▣ Under the Normal distribution both are $-\mu + \sigma \times$ constant: they add no information beyond volatility
- ▣ With heavy tails the constants change, and ES reacts more than VaR

Four Axioms of a Coherent Risk Measure

- A **risk measure** ρ maps a loss L to a number (capital needed) Artzner, Delbaen, Eber & Heath (1999)
- **Monotonicity**: $L_1 \leq L_2$ always $\Rightarrow \rho(L_1) \leq \rho(L_2)$
- **Translation invariance**: $\rho(L + c) = \rho(L) + c$ for a sure amount c
- **Positive homogeneity**: $\rho(\lambda L) = \lambda \rho(L)$ for $\lambda \geq 0$ (double the position, double the risk)
- **Subadditivity**: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$
 - ▶ “A merger does not create extra risk”: diversification never increases the measure

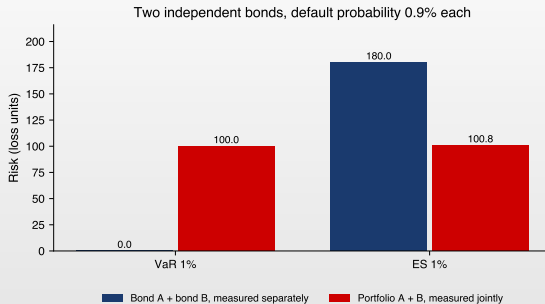
Why Subadditivity Matters

- **Limits:** a bank gives each desk a risk limit ρ_i
 - ▶ With subadditivity, the bank-wide risk is at most $\sum_i \rho_i$: decentralised control works
- **Regulatory arbitrage:** without it, splitting a firm into pieces could lower the capital requirement
- **Optimisation:** a subadditive, homogeneous measure is convex in the portfolio weights, so minimising it has a unique solution
- VaR satisfies the first three axioms, but **not** subadditivity in general

Counterexample: Two Bonds, Step by Step

- Bonds A and B, independent; each loses 100 with probability $p = 0.9\%$ (default) and 0 otherwise
- **One bond:** $P(X \leq -100) = p = 0.9\%$, not above 1%, so $q_{1\%} = 0$ and $\text{VaR}_{1\%}(A) = \text{VaR}_{1\%}(B) = 0$
 - ▶ $\text{ES}_{1\%}(A) = (100 \cdot 0.009 + 0 \cdot 0.001)/0.01 = 90$
- **Portfolio A + B:** $P(X \leq -100) = 1 - (1 - p)^2 = 1.79\% > 1\%$, $P(X \leq -200) = p^2 = 0.0081\%$, so $q_{1\%} = -100$
 - ▶ $\text{VaR}_{1\%}(A + B) = 100 > 0 + 0 = \text{VaR}_{1\%}(A) + \text{VaR}_{1\%}(B)$
 - ▶ General formula for a discrete loss $L = -X$: $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L \mathbf{1}\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha (\alpha - P(L > \text{VaR}_\alpha)))$
- $\text{ES}_{1\%}(A + B) = 100.81 \leq 180 = \text{ES}_{1\%}(A) + \text{ES}_{1\%}(B)$

Subadditivity: VaR Fails, ES Holds



□ Blue: sum of the two stand-alone measures; red: the measure of the combined portfolio

What the Counterexample Teaches

- VaR ignores everything beyond the quantile: each bond alone “hides” its default in the last 1%
 - ▶ Together, the chance of at least one default is $1.79\% > 1\%$ and VaR jumps to 100
- A VaR-based limit would reward **concentration**: holding only A looks riskless
- ES sees the size of the tail and is coherent [Acerbi & Tasche \(2002\)](#)
 - ▶ Also convex, so it can be minimised by linear programming [Rockafellar & Uryasev \(2002\)](#)
- Credit portfolios, options and other skewed pay-offs are where VaR fails in practice

When Is VaR Subadditive After All?

- **Elliptical distributions** (Normal, Student- t with common ν): VaR is subadditive [McNeil, Frey & Embrechts \(2015\)](#)
 - ▶ There VaR and ES are both $\mu_p + \sigma_p \times \text{constant}$, with $\sigma_p \leq \sum_i \sigma_i$
- **Heavy but not too heavy tails**: VaR is subadditive in the tail when the mean is finite (tail index above 1) [Dánielsson et al. \(2013\)](#)
 - ▶ Equity and FX returns usually satisfy this (tail index between 4.8 and 6.5 in Section 7)
- It fails for discrete jumps (defaults), extremely heavy tails and some option pay-offs
- VaR vs ES in practice: ES needs more data for the same precision [Yamai & Yoshida \(2005\)](#)

Preview: Elicitability (Chapter 8)

- A risk measure is **elicitable** if it is the minimiser of an expected scoring function
 - ▶ VaR: the quantile $q_\alpha(X)$ minimises the pinball loss $\mathbb{E}[(1\{X \leq v\} - \alpha)(v - X)]$
- ES alone is **not** elicitable [Gneiting \(2011\)](#): no score ranks ES forecasts correctly on its own
- But the pair (VaR, ES) is jointly elicitable [Fissler & Ziegel \(2016\)](#)
- Consequence: comparing ES forecasts is harder than comparing VaR forecasts
 - ▶ ES precision is governed by the tail sample $n\alpha$, not by n [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
 - ▶ S&P 500, 26 years: only 168 days in the tail of ES 2.5%

Entropy and Market Risk

- **Differential entropy** of a density f : $H(L) = - \int f(x) \ln f(x) dx$, a measure of dispersion without moments
 - ▶ For a location–scale family $L = \mu + \sigma Z$: $H(L) = \ln \sigma + H(Z)$, so $\text{VaR}_\alpha = \mu + e^{H(L)-H(Z)} q_\alpha(Z)$
- [Pele, Lazar & Dufour \(2017\)](#): the entropy of intraday returns is linked to intraday VaR and ES and helps forecast daily VaR (EUR/JPY)
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): high-frequency entropy gives the best daily VaR forecasts for Bitcoin among the models compared
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail-entropy measure as an alternative estimator of ES

Recap: Coherence

- ▣ Coherence = monotonicity, translation invariance, positive homogeneity and subadditivity
- ▣ VaR can violate subadditivity (the two-bond example); ES never does
- ▣ For elliptical and moderately heavy-tailed returns VaR is subadditive in practice
- ▣ VaR is elicitable, ES is not alone: this shapes backtesting in Chapter 8

Five Families of Methods

Method	Assumption	Weak point
Historical simulation	The past n days are a sample from tomorrow's distribution	Slow reaction; no loss beyond the worst seen
Parametric (Normal, t , Cornish–Fisher)	A distribution family with estimated parameters	Wrong family = wrong tail
Filtered historical simulation	GARCH volatility $\times$ empirical standardised residuals	Depends on the volatility model
Monte Carlo	Simulated paths from a fitted model	Model risk; computing time
Extreme value theory	Only the tail follows a limit law	Threshold choice; few data

- Every method gives VaR and ES from the same fitted distribution: ES_α is the average of VaR_u , $u < \alpha$

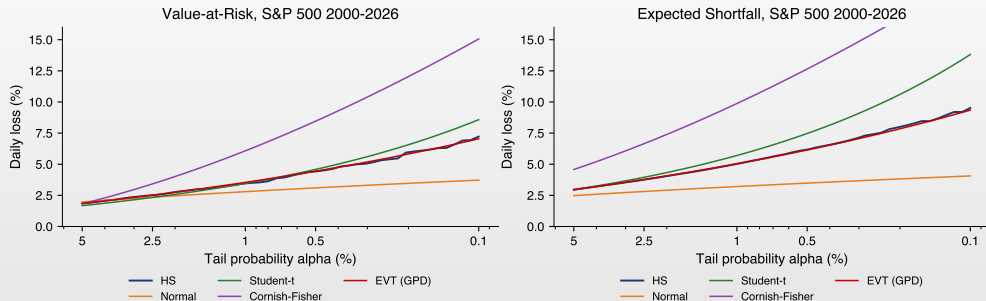
Historical Simulation

- Take the last n daily losses $L_{t-n+1}, \dots, L_t$ and sort them: $L_{(1)} \leq \dots \leq L_{(n)}$
 - ▶ $\widehat{\text{VaR}}_\alpha = L_{(\lceil n(1-\alpha) \rceil)}$ (or an interpolated empirical quantile); $\widehat{\text{ES}}_\alpha = \text{mean of the losses} \geq \widehat{\text{VaR}}_\alpha$
- No distributional assumption: heavy tails and skewness are kept automatically
- Choice of window n : 250 days reacts faster, 1000 days is more precise
 - ▶ With $n = 500$, VaR 1% rests on 5 observations and ES 2.5% on about 12
- **Ghost effect**: a crash day enters the window, stays for n days, then drops out and VaR falls abruptly

The Cornish–Fisher Expansion

- Adjust the Normal quantile for **skewness** $S = \mathbb{E}[(X - \mu)^3]/\sigma^3$ and **excess kurtosis** $K = \mathbb{E}[(X - \mu)^4]/\sigma^4 - 3$
Cornish & Fisher (1938)
 - ▶ $\tilde{z}_\alpha = z_\alpha + \frac{(z_\alpha^2 - 1)S}{6} + \frac{(z_\alpha^3 - 3z_\alpha)K}{24} - \frac{(2z_\alpha^3 - 5z_\alpha)S^2}{36}, \quad \text{VaR}_\alpha = -(\mu + \sigma\tilde{z}_\alpha), \text{ with } z_\alpha < 0$
- ES: average of $-(\mu + \sigma\tilde{z}_u)$ over $u \in (0, \alpha)$, numerically
- An expansion around the Normal distribution: accurate only for **small** S and K
- S&P 500 returns: $S = -0.35$, $K = 10.7$: far outside that range
 - ▶ $\text{VaR}_{1\%}^{CF} = 6.08\%$ versus historical 3.45%: the kurtosis term overshoots

Five Methods Across Tail Probabilities



□ S&P 500 daily losses 2000–2026; tail probabilities α from 5% down to 0.1%

Reading the Level Curves

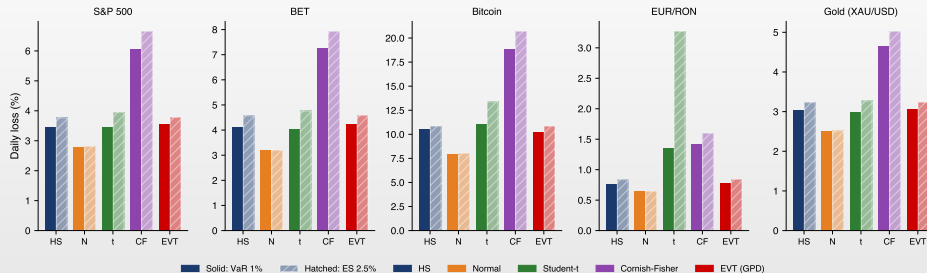
- VaR 0.1% (one day in four years): historical 7.22%, extreme-value 7.04%
 - ▶ Normal distribution: 3.72%, about half; Student- t : 8.58%; Cornish–Fisher: 15.06%
- The gap between methods **grows** as α falls: the further into the tail, the more the model matters
- ES curves spread out more than VaR curves: ES depends on the whole tail shape
- Extreme-value and historical estimates agree down to $\alpha = 0.5\%$; beyond it only extreme-value theory can extrapolate

VaR and ES by Method and Asset

	VaR _{1%} (%)					ES _{2.5%} (%)				
	Hist.	Normal	<i>t</i>	CF	EVT	Hist.	Normal	<i>t</i>	CF	EVT
S&P 500	3.45	2.80	3.46	6.08	3.54	3.76	2.81	3.95	6.65	3.78
BET	4.13	3.19	4.06	7.25	4.22	4.57	3.21	4.76	7.93	4.58
Bitcoin	10.51	7.99	11.06	18.89	10.23	10.84	8.03	13.37	20.69	10.84
EUR/RON	0.76	0.64	1.36	1.43	0.78	0.84	0.65	3.27	1.60	0.84
Gold	3.04	2.50	3.00	4.64	3.07	3.23	2.51	3.27	5.02	3.23

- Daily losses, full samples; CF = Cornish–Fisher, EVT = generalised Pareto tail fitted to the largest 5% of losses (Section 7)
- Bitcoin needs about three times the S&P 500 buffer; EUR/RON about a fifth

Methods Compared on Five Assets



□ Solid bars: $\text{VaR}_{1\%}$; hatched bars: $\text{ES}_{2.5\%}$; daily losses in %

What Goes Wrong, and Where

- **Normal distribution:** too low everywhere, e.g. Bitcoin $\text{VaR}_{1\%}$ 7.99% vs 10.51%
 - ▶ Its ES 2.5% is almost its VaR 1%: the Normal tail is thin
- **Cornish–Fisher:** excess kurtosis above 7 in every series makes it overshoot, e.g. BET 7.25% vs 4.13%
 - ▶ Kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0, EUR/RON 15.1, gold 7.7
- **Student- t :** good VaR, but EUR/RON gives $\hat{\nu} = 1.25$ (infinite variance) and an absurd ES of 3.27%
 - ▶ The peak of many near-zero reference-rate changes drives $\hat{\nu}$ down
- **Extreme value theory:** close to historical simulation for VaR 1% in all five series, with a smooth tail beyond

Filtered Historical Simulation

- Idea [Barone-Adesi, Giannopoulos & Vosper \(1999\)](#), [Hull & White \(1998\)](#): history gives the **shape** of the shocks, a GARCH model gives today's **scale**
 - ▶ $r_t = \mu_t + \sigma_t z_t$, $\mu_t = c + \phi r_{t-1}$, $\sigma_t^2 = \omega + \alpha_1 (r_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2$ (Chapter 5, [Bollerslev \(1986\)](#))
- Step 1: estimate the AR(1)-GARCH(1,1) model by quasi-maximum likelihood (Normal likelihood, robust to other shock distributions)
- Step 2: standardised residuals $\hat{z}_t = (r_t - \hat{\mu}_t) / \hat{\sigma}_t$: approximately i.i.d., still heavy-tailed
- Step 3: rescale by tomorrow's volatility
 - ▶ $\text{VaR}_{\alpha, t+1} = -(\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} q_\alpha(\hat{z}))$, $\text{ES}_{\alpha, t+1} = -\hat{\mu}_{t+1} - \hat{\sigma}_{t+1} \overline{(\hat{z} \mid \hat{z} \leq q_\alpha(\hat{z}))}$

GARCH Filters and Tomorrow's Volatility

	$\hat{\phi}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\alpha}_1 + \hat{\beta}_1$	$\hat{\sigma}$ (sample)	$\hat{\sigma}_{t+1}$	$-q_1\%(\hat{z})$
S&P 500	-0.052	0.120	0.860	0.981	1.21	0.73	2.79
BET	0.104	0.207	0.787	0.994	1.40	1.76	2.74
Bitcoin	-0.009	0.127	0.843	0.970	3.49	3.00	2.88
EUR/RON	0.169	0.200	0.780	0.980	0.28	0.11	2.74
Gold	-0.022	0.046	0.943	0.989	1.09	1.43	2.68

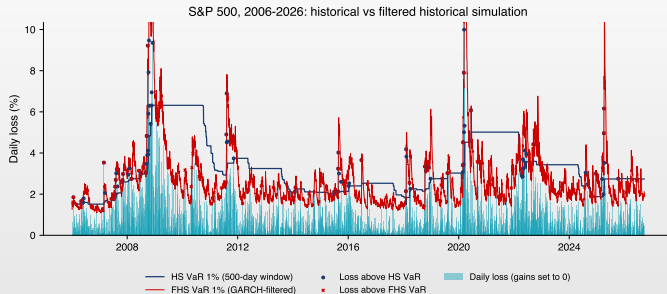
- $\hat{\sigma}_{t+1}$: forecast for the first trading day after 18 September 2026 (19 September for Bitcoin), in %
- Persistence $\hat{\alpha}_1 + \hat{\beta}_1$ close to 1: volatility shocks die out slowly (Chapter 5)
- S&P 500 and EUR/RON are calm today ($\hat{\sigma}_{t+1}$ below the sample value); BET and gold are turbulent
- $-q_1\%(\hat{z})$ is between 2.68 and 2.88, above $-z_1\% = 2.326$: the filtered shocks are still heavy-tailed

Conditional VaR and ES for the Next Trading Day

	VaR _{1%} (%)				ES _{1%} (%)			
	GARCH	FHS	c-EVT	Hist.	GARCH	FHS	c-EVT	Hist.
S&P 500	1.63	1.97	2.03	3.45	1.88	2.55	2.56	5.01
BET	3.80	4.53	4.63	4.13	4.39	6.16	6.20	6.28
Bitcoin	6.89	8.56	8.68	10.51	7.90	12.22	12.19	14.29
EUR/RON	0.23	0.27	0.26	0.76	0.27	0.34	0.35	1.12
Gold	3.32	3.82	3.86	3.04	3.80	4.94	4.93	4.21

- GARCH: GARCH with Normal shocks; c-EVT: conditional extreme value theory (Section 7); Hist.: unconditional, full sample
- Conditional numbers move with the market: S&P 500 FHS VaR 1.97% today vs 3.45% unconditional
- GARCH with Normal shocks is the lowest of the three conditional methods in every row: volatility alone does not create the tail

Historical vs Filtered Historical Simulation: S&P 500

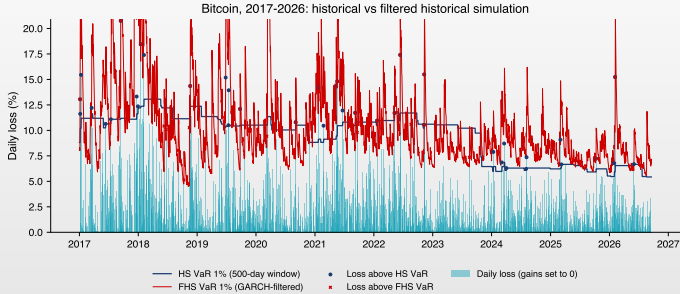


- Daily one-day-ahead $\text{VaR}_{1\%}$; GARCH parameters re-estimated each January on all earlier data, so no future information is used

Reading the S&P 500 Comparison

- ▣ Share of days with a loss above VaR 1%, 2006–2026 (5 208 days; target $\alpha = 1\%$)
 - ▶ Historical, 500 days: 1.67%; filtered: 1.54%; GARCH with Normal shocks: 2.34%
- ▣ Historical VaR moves between 1.49% and 6.32%; filtered VaR between 1.09% and 18.02% (peak on 2020-03-18)
- ▣ Historical VaR exceedances come in **clusters** (2008, 2020): the window reacts months too late
- ▣ Filtered VaR spreads exceedances over time: what a good forecast should do (formal tests in Chapter 8)

Historical vs Filtered Historical Simulation: Bitcoin



- Same design as for the S&P 500; forecasts from January 2017, 7 days a week

Reading the Bitcoin Comparison

- ▣ Share of days above VaR (3 548 days): historical 1.16%, filtered 0.76%, GARCH with Normal shocks 1.58%
- ▣ Filtered VaR peaked at 54.83% in March 2020, just after the crash of 2020-03-12 (46.47% loss in one day)
- ▣ Bitcoin volatility has fallen since 2022; the 500-day historical VaR follows with a long delay
- ▣ Crypto as a distinct asset class with its own risk profile: [Pele et al. \(2023\)](#)

Monte Carlo Simulation

- Simulate many future scenarios from a fitted model; read VaR and ES from the simulated losses
 - ▶ Needed when the loss is a non-linear function of the risk factors (options) or the horizon is several days
- **Filtered Monte Carlo** for h days: repeat for $j = 1, \dots, M$
 - ▶ draw $z_{t+1}^{(j)}, \dots, z_{t+h}^{(j)}$ with replacement from the standardised residuals $\hat{z}$
 - ▶ update $\sigma^{(j)}$ and $r^{(j)}$ day by day with the GARCH recursion; $L^{(j)} = -\sum_{i=1}^h r_{t+i}^{(j)}$
- $\widehat{\text{VaR}}_\alpha, \widehat{\text{ES}}_\alpha =$ historical estimates on the M simulated losses; simulation error falls as $1/\sqrt{M}$
- Keeps volatility clustering, heavy tails and the mean reversion of volatility over the horizon

Recap: Methods

- ▣ Historical simulation: simple and model-free, but slow and blind beyond the worst observed loss
- ▣ Normal distribution: underestimates the tail; Cornish–Fisher: unreliable with the kurtosis of daily returns
- ▣ Student- t : a good one-parameter tail, fragile when $\hat{\nu}$ approaches 2
- ▣ Filtered historical simulation and Monte Carlo: today's volatility + empirical shock shape; our default for daily risk

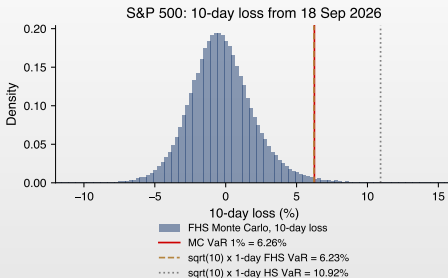
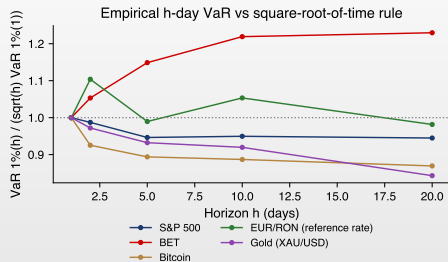
The Square-Root-of-Time Rule

- If daily losses are i.i.d. $N(0, \sigma^2)$, the h -day loss is $N(0, h\sigma^2)$
 - ▶ $\text{VaR}_\alpha^{(h)} = \sqrt{h} \text{VaR}_\alpha^{(1)}$, $\text{ES}_\alpha^{(h)} = \sqrt{h} \text{ES}_\alpha^{(1)}$
- Ten days: multiply by $\sqrt{10} = 3.162$; Basel 1996 allowed exactly this scaling of one-day VaR 1% [Basel Committee \(1996\)](#)
- The rule needs **three** assumptions: zero mean, no autocorrelation, constant volatility with Normal shocks
- Each assumption fails in the data (Chapters 1, 2 and 5)

When the Rule Fails

- **Autocorrelation:** $\text{Var}(r_{t+1} + \dots + r_{t+h}) = h\sigma^2 \text{VR}(h)$ with the variance ratio of Chapter 2
 - ▶ Positive autocorrelation ($\text{VR} > 1$): $\sqrt{h}$ **underestimates** the h -day risk
- **Volatility clustering:** over h days volatility reverts to its mean
 - ▶ Calm today: $\sqrt{h}$ underestimates; turbulent today: it overestimates
- **Heavy tails:** sums of heavy-tailed shocks are closer to the Normal distribution (Chapter 1)
 - ▶ For i.i.d. shocks with tail index $a > 2$ the far-tail quantile of a sum grows like $h^{1/a} < \sqrt{h}$ [Dánielsson \(2011\)](#)
- With jumps, the rule can badly misstate multi-day risk [Dánielsson & Zigrand \(2006\)](#)

Square-Root-of-Time in the Data



- Left: empirical $\text{VaR}_{1\%}$ of overlapping h -day losses divided by $\sqrt{h}$ times the one-day value; right: filtered Monte Carlo, 200 000 paths

Reading the Horizon Chart

- Ratio at $h = 10$: BET 1.22 (lag-1 autocorrelation 0.109), EUR/RON 1.05 (0.170)
 - ▶ S&P 500 0.95 (-0.099), gold 0.92, Bitcoin 0.89: heavy tails and reversal make $\sqrt{h}$ conservative
- S&P 500 today: $\hat{\sigma}_{t+1} = 0.73\%$ vs 1.21% in the sample
 - ▶ 10-day $\text{VaR}_{1\%}$: filtered Monte Carlo 6.26% ; $\sqrt{10} \times$ one-day filtered 6.23% ; $\sqrt{10} \times$ one-day historical 10.92%
- The mean reversion of volatility and the heavy tails of the shocks roughly offset each other here
- The big difference is conditional vs unconditional, not $\sqrt{h}$ vs simulation

Recap: Horizon

- $\sqrt{h}$ is exact only for i.i.d. Normal losses with zero mean
- Positive autocorrelation (BET) makes it too low; heavy tails and reversal (S&P 500, Bitcoin) too high
- Simulation from a filtered model handles all effects at once
- Direct h -day historical estimates use overlapping sums: fewer independent observations

Portfolio VaR under the Normal Distribution

- Weights $w = (w_1, \dots, w_d)$, simple returns R with covariance matrix Σ ; portfolio loss $L_p = -w^\top R$
 - ▶ $\sigma_p = \sqrt{w^\top \Sigma w}$, $\text{VaR}_\alpha(L_p) = z_{1-\alpha} \sigma_p$ (mean ignored at one day)
- Simple returns add across assets, log returns add over time: portfolios use simple returns
- **Undiversified VaR**: $\sum_i z_{1-\alpha} w_i \sigma_i \geq \text{VaR}_\alpha(L_p)$, equality only if all correlations are 1
 - ▶ The difference is the **diversification benefit**
- Question for the risk manager: which position should be cut to reduce risk most?

Marginal and Component VaR: Euler Allocation

- **Marginal VaR:** sensitivity to a small increase in weight i

►
$$\frac{\partial \text{VaR}_\alpha}{\partial w_i} = \mathbf{z}_{1-\alpha} \frac{(\Sigma \mathbf{w})_i}{\sigma_p}$$

- **Component VaR:** $C_i = w_i \partial \text{VaR}_\alpha / \partial w_i$

► Euler's theorem: a function homogeneous of degree 1 satisfies $\sum_i w_i \partial \rho / \partial w_i = \rho$, so $\sum_i C_i = \text{VaR}_\alpha$ [Tasche \(2008\)](#)

- The same principle for ES without any distributional assumption

► $C_i^{ES} = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha(L_p)]$, $\sum_i C_i^{ES} = \text{ES}_\alpha(L_p)$; historical estimate: average over the tail days

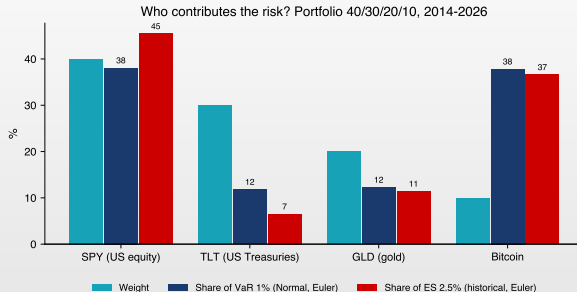
- A negative C_i means the asset **hedges** the portfolio

The Portfolio: SPY, TLT, GLD and Bitcoin

	Weight	σ_i (%)	corr. SPY	corr. TLT	corr. GLD
SPY	40%	1.10	1.00		
TLT	30%	0.93	-0.17	1.00	
GLD	20%	1.03	0.07	0.25	1.00
Bitcoin	10%	4.14	0.23	-0.02	0.09

- Daily simple returns on 3017 common trading days, September 2014 – September 2026; daily rebalancing
- Portfolio $\sigma_p = 0.76\%$; $\text{VaR}_{1\%}$ Normal = 1.77%, undiversified 3.11%: diversification removes 43%
- Historical $\text{ES}_{2.5\%} = 2.15\%$ vs undiversified 3.87% (44% less)

Who Contributes the Risk?



□ Teal: weights; blue: shares of Normal $\text{VaR}_{1\%}$; red: shares of historical $\text{ES}_{2.5\%}$ (Euler allocation)

Reading the Risk Contributions

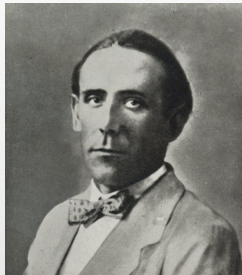
	Marginal VaR	C_i	share	C_i^{ES}	share
SPY	1.68	0.67	38.1%	0.98	45.4%
TLT	0.70	0.21	11.8%	0.14	6.5%
GLD	1.08	0.22	12.3%	0.25	11.4%
Bitcoin	6.68	0.67	37.8%	0.79	36.6%
Total		1.77	100%	2.15	100%

- Bitcoin: 10% of the money, 37.8% of VaR and 36.6% of ES: its volatility is almost four times that of SPY
- TLT: 30% of the money, only 6.5% of ES: on the portfolio's worst days TLT lost only 0.47% on average
- SPY carries more of the ES than of the Normal VaR (45.4% vs 38.1%): tail dependence is stronger than correlation

Recap: Portfolio Risk

- Portfolio VaR is below the sum of stand-alone VaRs whenever correlations are below 1
- Euler allocation splits VaR or ES into contributions that add up exactly
- Risk shares, not money weights, show where the risk is: small positions can dominate
- Historical ES contributions capture the dependence in crashes, which correlations miss (Chapter 6)

Emil Gumbel and the Statistics of Extremes



Emil Julius Gumbel (1891–1966)

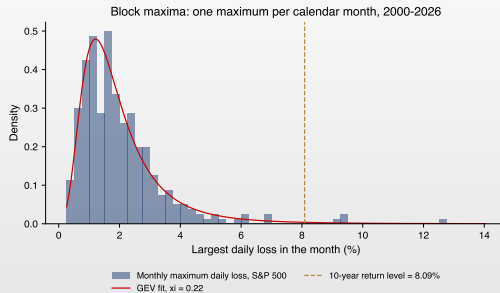
Photo: unknown author (c. 1931); public domain;
Wikimedia Commons

- Fisher & Tippett (1928): the maximum of many i.i.d. variables, suitably rescaled, can have only three limit forms
- Gnedenko (1943) proved the result rigorously
- Gumbel (1958) wrote the first textbook and applied it to floods, droughts and life lengths
 - ▶ The Gumbel distribution is one of the three limit forms
- In finance: model **only the tail**, where the risk is Embrechts, Klüppelberg & Mikosch (1997), Longin (2000)

Block Maxima and the GEV Distribution

- Split the sample into blocks of n days; keep each block maximum $M_n = \max(L_1, \dots, L_n)$
 - ▶ Fisher–Tippett–Gnedenko theorem: if $(M_n - b_n)/a_n$ has a non-degenerate limit, the limit is GEV
- $H_\xi(x) = \exp \left\{ - \left(1 + \xi \frac{x-\mu}{\sigma} \right)^{-1/\xi} \right\}$, $1 + \xi(x - \mu)/\sigma > 0$ ($\xi = 0$: $\exp\{-e^{-(x-\mu)/\sigma}\}$)
- The **shape** ξ decides the tail
 - ▶ $\xi > 0$: Fréchet, power-law tail with tail index $1/\xi$ (financial returns)
 - ▶ $\xi = 0$: Gumbel, exponential tail (Normal distribution); $\xi < 0$: Weibull, bounded tail
- **Return level** for k blocks: the level exceeded on average once every k blocks, $H_\xi^{-1}(1 - 1/k)$

Monthly Maxima of S&P 500 Losses



- 321 calendar months, 2000–2026; GEV fitted by maximum likelihood

Reading the Block Maxima

- $\hat{\xi} = 0.22$ (Fréchet), $\hat{\mu} = 1.37$, $\hat{\sigma} = 0.79$: a power-law tail with index about $1/\hat{\xi}$
 - ▶ 10-year return level: 8.09%; 30-year: 10.95%; the largest monthly maximum was 12.77%
- From monthly maxima to a daily VaR: $P(M_n \leq x) \approx P(L \leq x)^n$, so $\text{VaR}_\alpha \approx H^{-1}((1 - \alpha)^n)$ with $n = 20.9$
 - ▶ $\text{VaR}_{1\%} \approx H^{-1}(0.810) = 2.83\%$ vs historical 3.45%; $\text{VaR}_{0.1\%} \approx 6.19\%$
- The daily approximation assumes independent days; volatility clustering breaks it, and it is too low here
- Block maxima throw away data (one value per month): the next method uses all large losses

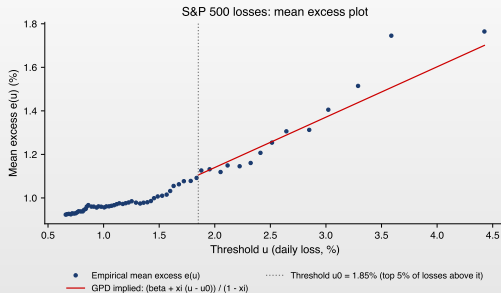
Peaks over Threshold and the GPD

- Fix a high threshold u ; model the **excesses** $Y = L - u$ given $L > u$
 - ▶ Pickands–Balkema–de Haan theorem [Pickands \(1975\)](#), [Balkema & de Haan \(1974\)](#): for a wide class of distributions, as u grows the excesses follow a generalised Pareto distribution
- $G_{\xi, \beta}(y) = 1 - (1 + \xi y / \beta)^{-1/\xi}$, $y \geq 0$, scale $\beta > 0$; the same ξ as the GEV
- Tail estimator with N_u exceedances among n observations [McNeil, Frey & Embrechts \(2015\)](#)
 - ▶ $\widehat{\text{VaR}}_\alpha = u + \frac{\hat{\beta}}{\hat{\xi}} \left[\left(\frac{n}{N_u} \alpha \right)^{-\hat{\xi}} - 1 \right]$, $\widehat{\text{ES}}_\alpha = \frac{\widehat{\text{VaR}}_\alpha}{1 - \hat{\xi}} + \frac{\hat{\beta} - \hat{\xi}u}{1 - \hat{\xi}}$ ($\hat{\xi} < 1$)
- Bias–variance trade-off: a low u violates the limit theorem, a high u leaves few exceedances

Choosing the Threshold: the Mean Excess Function

- ▣ **Mean excess function:** $e(u) = \mathbb{E}[L - u \mid L > u]$
 - ▶ Empirical version: average of the excesses above u , plotted against u
- ▣ For a GPD tail $e(u)$ is **linear** in u
 - ▶ $e(u) = \frac{\beta + \xi(u - u_0)}{1 - \xi}$, slope $\xi/(1 - \xi)$: upward for heavy tails, flat for exponential tails
- ▣ Rule of thumb: choose u where the plot becomes roughly linear; check the stability of $\hat{\xi}$ across thresholds
- ▣ The Hill estimator of Chapter 1 [Hill \(1975\)](#) estimates the same tail index $1/\xi$

Mean Excess Plot: S&P 500

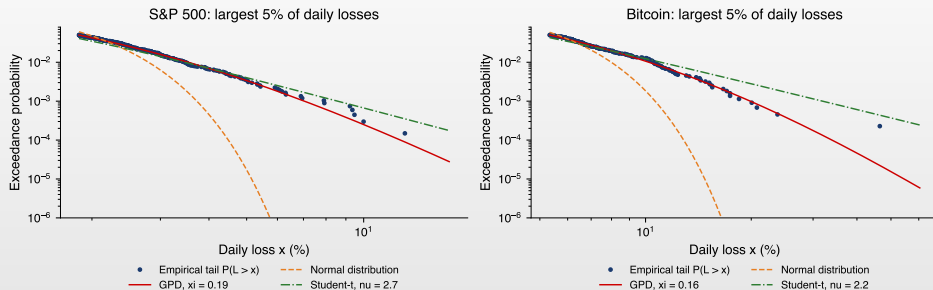


- Thresholds leaving from 20% down to 0.5% of daily losses above them; red line: GPD fitted to the largest 5%

Reading the Mean Excess Plot

- ▣ The empirical mean excess rises with u : a heavy (Fréchet-type) tail, not the flat line of an exponential tail
- ▣ At $u_0 = 1.85\%$ (top 5% of losses above it) the mean excess is 1.09%; the GPD slope $\hat{\xi}/(1 - \hat{\xi}) = 0.23$ follows the points
- ▣ Beyond VaR 1% the points scatter: each is an average of very few losses
- ▣ Keeping the top 5% leaves $N_u = 336$ exceedances: enough for a stable fit

GPD Tails: S&P 500 and Bitcoin



□ Log-log scale: probability of a daily loss above x , for the largest 5% of losses

MFM_ch7_gpd_tail

Reading the GPD Fits

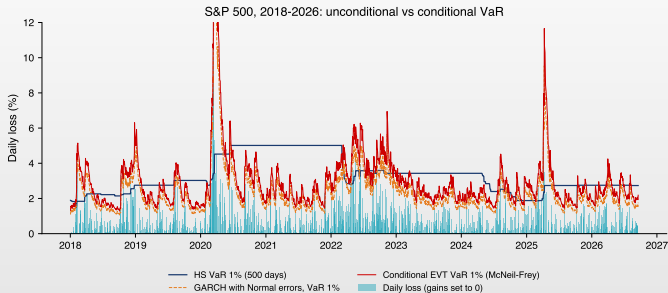
	u (%)	N_u	$\hat{\xi}$	$SE(\hat{\xi})$	$1/\hat{\xi}$	$VaR_{1\%}$	$ES_{1\%}$
S&P 500	1.85	336	0.19	0.065	5.3	3.54	5.04
BET	1.93	334	0.21	0.066	4.8	4.22	6.34
Bitcoin	5.33	220	0.16	0.078	6.1	10.23	14.36
EUR/RON	0.38	268	0.18	0.072	5.5	0.78	1.14
Gold	1.68	348	0.15	0.062	6.5	3.07	4.22

- $\hat{\xi}$ between 0.15 and 0.21 for all five series, each more than two standard errors above 0: heavy tails everywhere
- Tail index $1/\hat{\xi}$ between 4.8 and 6.5: moments up to order four exist (compare Chapter 1)
- On the chart the Normal tail falls far below the data; the Student- t tail is close to the GPD

Conditional EVT (McNeil and Frey)

- Unconditional EVT ignores volatility clustering; [McNeil & Frey \(2000\)](#) combine GARCH filtering with a GPD tail
 - ▶ Step 1: fit AR(1)-GARCH(1,1) by quasi-maximum likelihood; standardised residuals $\hat{z}_t$
 - ▶ Step 2: fit a GPD to the largest 10% of $-\hat{z}_t$ and compute its VaR $\hat{z}_\alpha^{GPD}$ and ES at level α
 - ▶ Step 3: $\text{VaR}_{\alpha, t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{z}_\alpha^{GPD}$, with $\hat{z}_\alpha^{GPD}$ the GPD VaR of $-\hat{z}$; ES in the same way
- Same as filtered historical simulation, but with a smooth parametric tail beyond the data
- Matters most far in the tail (VaR 0.5%, VaR 0.1%), where only a handful of residuals exist

Conditional EVT on the S&P 500



□ One-day-ahead $\text{VaR}_{1\%}$, 2018–2026; parameters re-estimated each January on all earlier data

Reading the Conditional EVT Chart

- ▣ Share of days above VaR 1%, 2018–2026 (2 190 days; target $\alpha = 1\%$)
 - ▶ Conditional EVT 1.23%; GARCH with Normal shocks 2.42%; historical, 500 days 1.60%
- ▣ The GARCH–Normal line has the right **timing** but the wrong **level**: its exceedance rate is more than twice the target
- ▣ Conditional EVT keeps the timing and fixes the level: the tail of the shocks, not only their scale
- ▣ Whether 1.2% is statistically different from 1% is a question for Chapter 8

Recap: Extreme Value Theory

- ▣ Two limit theorems: block maxima $\rightarrow$ GEV, excesses over a high threshold $\rightarrow$ GPD; both governed by ξ
- ▣ All five series: $\hat{\xi} \approx 0.15\text{--}0.22$, a Fréchet-type heavy tail
- ▣ Peaks over threshold uses the data better than block maxima; the mean excess plot guides the threshold
- ▣ Conditional EVT = GARCH scale + GPD tail: the most complete one-day model in this chapter

Liquidity-Adjusted VaR

- VaR assumes positions can be sold at the mid price; in reality we pay half the bid–ask spread
 - ▶ **Relative spread** $S = (\text{ask} - \text{bid})/\text{mid}$; it widens exactly when markets fall
- [Bangia, Diebold, Schuermann & Stroughair \(2001\)](#): $\text{LVaR}_\alpha = \text{VaR}_\alpha + \frac{1}{2}(\mu_S + a\sigma_S)$, with mean μ_S and standard deviation σ_S of the spread and a multiplier a
- Example: Banca Transilvania, 1 million RON; historical $\text{VaR}_{1\%} = 3.21\%$ over the last two years (496 days)
 - ▶ Assumed spread: $\mu_S = 0.5\%$, $\sigma_S = 0.3\%$, $a = 3$; cost of liquidation 0.70%
 - ▶ $\text{LVaR} = 3.91\%$: 39 090 RON instead of 32 090 RON (+22%)
- Large positions also move the price (endogenous liquidity): the regulator handles this with longer horizons

The Basel Committee



The BIS tower in Basel, seat of the Basel Committee on Banking Supervision

Photo: Michal Pleskowicz (2018); CC BY-SA 4.0;
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- The Basel Committee on Banking Supervision sets global capital standards; national supervisors (in the EU, via the Capital Requirements Regulation) turn them into law
- **1996:** market risk enters the capital rules; banks may use internal VaR models [Basel Committee \(1996\)](#)
- **2019:** the Fundamental Review of the Trading Book replaces VaR by ES [Basel Committee \(2019\)](#)

Basel 1996: VaR 1% over 10 Days

- ▣ VaR 1% (one-tailed), computed daily, minimum holding period of 10 trading days [Basel Committee \(1996\)](#)
- ▣ One-day VaR may be scaled to 10 days by the square root of time
- ▣ At least one year of historical data
- ▣ Capital: the larger of yesterday's VaR and the 60-day average VaR times a multiplication factor of at least 3
 - ▶ A “plus” of 0 to 1 is added when backtesting finds too many exceptions (Chapter 8)
- ▣ Critique after 2008: VaR ignores the size of tail losses, and one-year windows forget crises

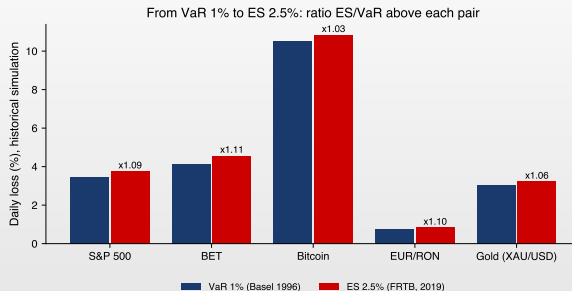
FRTB (2019): Expected Shortfall 2.5%

- ▣ ES 2.5% (one-tailed), computed daily for every trading desk using internal models [Basel Committee \(2019\)](#)
- ▣ Base horizon of 10 days, scaled up by **liquidity horizons** per risk factor
 - ▶ Large-cap equity 10 days, small-cap equity 20 days, specified currency pairs 10 days, credit spreads up to 120 days
- ▣ Calibrated to the most severe 12-month period of stress available
- ▣ Capital multiplier $m_c = 1.5$, increased by supervisors after poor backtesting
- ▣ Desks are still backtested on one-day VaR 2.5% and VaR 1%: VaR stays, ES sets the capital

Why ES 2.5% Replaces VaR 1%

- Under the Normal distribution the two are almost equal
 - ▶ $ES_{2.5\%} = \sigma \times 2.338$ vs $VaR_{1\%} = \sigma \times 2.326$: the same capital for thin tails
- With heavy tails $ES_{2.5\%} > VaR_{1\%}$: the new rule charges more exactly where the tail is heavy
- ES is coherent, so desk-level ES limits add up safely
- Price: ES is harder to backtest (not elicitable alone) and less precise than VaR for the same sample

VaR 1% vs ES 2.5% on Five Assets



- Historical simulation, daily losses, full samples; the number above each pair is $ES_{2.5\%}/VaR_{1\%}$

Reading the Basel Comparison

- $ES_{2.5\%}/VaR_{1\%}$: S&P 500 1.09, BET 1.11, Bitcoin 1.03, EUR/RON 1.10, gold 1.06
 - ▶ Normal distribution: $2.338/2.326 \approx 1.005$
- All five ratios are above the Normal value: the switch raises capital by 3–11% for the same positions
- The increase is largest for BET and EUR/RON, whose tails beyond VaR 1% are longest relative to VaR itself
- Regulatory capital also adds the stressed calibration and liquidity horizons: see Seminar 7, Part C

Recap: Liquidity and Regulation

- ▣ Liquidity-adjusted VaR adds half the (stressed) spread to the market VaR
- ▣ Basel 1996: VaR 1% over 10 days, $\sqrt{10}$ scaling allowed, multiplier at least 3
- ▣ FRTB 2019: ES 2.5%, 10-day base horizon, liquidity horizons, stressed calibration, multiplier 1.5
- ▣ VaR remains the backtesting tool; ES sets the capital

Summary

- ▣ VaR is a quantile of the loss, ES the average beyond it; ES is coherent, VaR is not in general
- ▣ Heavy tails: the Normal distribution underestimates, Cornish–Fisher overshoots, Student- t and EVT fit well
- ▣ Volatility clustering: filtered historical simulation and conditional EVT react in days, not months
- ▣ $\sqrt{h}$ scaling fails with autocorrelation, clustering and heavy tails; simulate instead
- ▣ Euler allocation shows where the risk sits; Basel now sets capital by ES 2.5%

Which Method When?

Situation	Suggested method
Quick check, long stable history	Historical simulation, 500–1000 days
Daily risk after a volatility shock	Filtered historical simulation or conditional EVT
VaR 0.5% and deeper in the tail, stress tests	Peaks over threshold (GPD), conditional or not
Multi-day horizon, options, non-linear pay-offs	Filtered Monte Carlo
Risk budgeting across desks or assets	Euler contributions to ES
Communication with a regulator	ES 2.5%, 10 days, stressed period, plus VaR 1% backtests

Coming Next: Backtesting (Chapter 8)

- ▣ How often did losses exceed VaR, and were the exceedances independent?
- ▣ Unconditional and conditional coverage tests; the Basel traffic-light zones
- ▣ Backtesting ES: joint scoring functions for (VaR, ES) Fissler & Ziegel (2016)
- ▣ Further reading: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), McNeil, Frey & Embrechts (2015)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07

References (1/3)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Balkema, A.A., de Haan, L. (1974). [Residual life time at great age](#). *Annals of Probability*, 2(5), 792–804.
- ▣ Bangia, A., Diebold, F.X., Schuermann, T., Stroughair, J.D. (2001). [Modeling liquidity risk, with implications for traditional market risk measurement and management](#). In *Risk Management: The State of the Art*, Springer, 3–13.
- ▣ Barone-Adesi, G., Giannopoulos, K., Vosper, L. (1999). [VaR without correlations for portfolios of derivative securities](#). *Journal of Futures Markets*, 19(5), 583–602.
- ▣ Basel Committee on Banking Supervision (1996). [Amendment to the capital accord to incorporate market risks](#). Bank for International Settlements.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#). Bank for International Settlements.
- ▣ Bollerslev, T. (1986). [Generalized autoregressive conditional heteroskedasticity](#). *Journal of Econometrics*, 31(3), 307–327.
- ▣ Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). [Statistical Tools for Finance and Insurance](#), 2nd ed. Springer.
- ▣ Cornish, E.A., Fisher, R.A. (1938). [Moments and cumulants in the specification of distributions](#). *Revue de l'Institut International de Statistique*, 5(4), 307–320.
- ▣ Daniélsson, J. (2011). [Financial Risk Forecasting](#). Wiley.
- ▣ Daniélsson, J., Jorgensen, B.N., Samorodnitsky, G., Sarma, M., de Vries, C.G. (2013). [Fat tails, VaR and subadditivity](#). *Journal of Econometrics*, 172(2), 283–291.

References (2/3)

- ▣ Daniélsso, J., Zigrand, J.-P. (2006). On time-scaling of risk and the square-root-of-time rule. *Journal of Banking & Finance*, 30(10), 2701–2713.
- ▣ Embrechts, P., Klüppelberg, C., Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- ▣ Fisher, R.A., Tippett, L.H.C. (1928). Limiting forms of the frequency distribution of the largest or smallest member of a sample. *Mathematical Proceedings of the Cambridge Philosophical Society*, 24(2), 180–190.
- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Gnedenko, B. (1943). Sur la distribution limite du terme maximum d'une série aléatoire. *Annals of Mathematics*, 44(3), 423–453.
- ▣ Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Gumbel, E.J. (1958). *Statistics of Extremes*. Columbia University Press.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Hull, J., White, A. (1998). Incorporating volatility updating into the historical simulation method for value-at-risk. *Journal of Risk*, 1(1), 5–19.
- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document*, 4th ed. New York.

References (3/3)

- ▣ Longin, F.M. (2000). From value at risk to stress testing: the extreme value approach. *Journal of Banking & Finance*, 24(7), 1097–1130.
- ▣ McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). Modeling expected shortfall using tail entropy. *Entropy*, 21(12), 1204.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiaty, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Pickands, J. (1975). Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1), 119–131.
- ▣ Rockafellar, R.T., Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7), 1443–1471.
- ▣ Tasche, D. (2008). Capital allocation to business units and sub-portfolios: the Euler principle. arXiv:0708.2542.
- ▣ Yamai, Y., Yoshida, T. (2005). Value-at-risk versus expected shortfall: a practical perspective. *Journal of Banking & Finance*, 29(4), 997–1015.